A Retrospective on Five Studies of LLM-Based Driving Evaluation

Six Cases Where the Headline Number Was Decided by Something Other Than the Model

Abstract

A language or vision-language model that scores, ranks or adjudicates is now a standard component of evaluation pipelines. Its reported quality is almost always a single headline number. We report six instances, measured across 4 of the 5 driving-evaluation studies run by one group under a common protocol, in which that headline number was decided by a variable unrelated to the capability under test. In each instance the mechanism was identified and quantified, not conjectured: a majority-class predictor that beat 4 of the 5 model arms; a model-free executor of the program already printed in the prompt that scored 24/24 and 12/12 with 0 model calls; a training-seed noise floor that reached 0.1574 while the three noise anchors used to size the experiment were 0.0121, 0.0435, 0.0652; 7 semantics-preserving rewordings of a single clause that moved a zero-shot endpoint across a span of 0.0271, wider than the reported margin of +0.0250, in a setting that was byte-identical across reruns; a gate criterion that read 0.8947 for all 3 arms including a constant-velocity control, because it was a property of the data pipeline; and one comparison that reversed sign between two rounds because the reference arm had become stronger. The last of the six cuts the other way: an endpoint sitting at 0.5250 against a trivial baseline of 0.5000 looked like absent capability, but ranking the same decisions by the probability the model placed on the token it had already emitted gave an AURC of 0.3392 against 0.4750 for random ordering, and thresholding that probability transferred to non-overlapping scenarios, moving 0.5150 to 0.6650. We draw a checklist from the six mechanisms, report it as a reusable scorer, and state plainly what this evidence cannot support.

1 Introduction

Evaluating a model is increasingly done by another model. Judges score answers, critics rank candidate actions, and reward models stand in for human preference. When such a system is proposed, its quality is reported as a number: an accuracy, an AUC, a win rate against a prior judge. The reader is then asked to treat that number as a measurement of the judge's competence.

This report describes six occasions on which that reading would have been wrong, and on which we were able to say what the number was actually measuring. The instances come from 4 of 5 studies in one domain, autonomous-driving evaluation, run by one group over roughly a year under a shared protocol. Each study had its own preregistration, its own frozen analysis script and its own held-out split. In each, a headline number was produced, written down, and only afterwards found to be controlled by a variable that had nothing to do with the capability the study was about.

We stress the chronology because it is the uncomfortable part. These are not failures we designed an experiment to expose. They are failures that survived preregistration, survived a frozen analysis, and were caught later—four of the six only after a reviewer asked for a baseline that had not been reported. The question this report is organised around is therefore not whether careful protocol prevents the problem. It did not. The question is which specific checks would have caught each case, and what they cost.

What we claim. Six mechanisms, each measured and each traceable to a stored artifact:

1. A **trivial upper bound** on the metric. On the layer where the study made its central claim, the majority-class predictor scored 0.7031 and 4 of the 5 rescored arms came in below it (§4). In a second study, a model-free executor of the decision program that the prompt already contained scored 24/24 and 12/12 using 0 model calls.
2. A **training-seed noise floor** larger than the effects being compared. Holding the arm fixed and changing only the training seed moved a safety endpoint by up to 0.1574; 2 of the 3 preregistered contrasts changed sign between the two seeds (§5).
3. A **prompt-wording noise floor** in an evaluation with no training at all. The same prompt, run twice, gave byte-identical answers; 7 semantics-preserving rewordings of one clause moved the primary endpoint across a span of 0.0271, against a reported margin of +0.0250 (§6).
4. An **arm-independent quantity** standing in a gate. All 3 arms, including a constant-velocity control, failed the same criterion at the same value, 0.8947, because the criterion measured the data pipeline rather than the arm (§7).
5. A **reference arm whose strength decided the answer**. The same comparison, run twice with the same budget rule, reversed sign because the baseline it was compared against had improved (§8).
6. A **readout gap** that is the opposite failure, and the one case where the pessimistic reading was wrong. An endpoint at 0.5250 against a trivial baseline of 0.5000 suggested the capability was absent; the probability the model had placed on the token it already emitted ranked the same decisions at AURC 0.3392 against 0.4750, and a threshold on it transferred to scenarios disjoint from the fitting set (§9).

Scope. Every instance is a statement about a measurement, not about whether models can judge driving. The six come

from one group and one toolchain, which bounds what they can establish: they are existence proofs with a mechanism attached to each, not an estimate of how often such failures occur. §12 lists the specific limits, including one case where this report's own subject recurred inside it.

2 The programme

Table 1 is the map. 5 studies ran over roughly a year, each with its own preregistration, frozen analysis script and held-out split, all aimed at one application: deciding whether a logged driving trajectory violates a rule. six instances in this report come from the 4 studies that produced quantitative endpoints.

Why the studies arose in this order. The first question was the obvious one: can a model execute rules that a Python evaluator already executes, and does it help to supervise the derivation rather than only the verdict? That study produced an affirmative answer on the easy layer and a negative one on the hard layer, and in doing so produced the two measurement failures of §4 and §5.

Each later study began where an earlier one stopped. Rule execution assumed the rule was known and applicable; asking what happens when the evidence is incomplete became the evidence-sufficiency study, which is where the wording effect (§6) and the readout gap (§9) were found. Rule execution also assumed a recording to evaluate; asking whether a learned world model can stand in for one became the replay study, whose gate turned out to measure the pipeline (§7). The question of whether a rule can be supplied at inference time instead of trained in became the rule-card study, which supplied the model-free baseline in §4. The fifth study compared general multimodal reasoners against a frozen text baseline on driving clips; with a single clip it produced no endpoint, but its result—that fluent narration replaced uncertainty, and that the larger model produced timestamps the frames did not support—is why uncertainty preservation is now checked as its own column rather than inferred from how confident an answer reads.

What was being built. The intended artifact throughout was a judge: a model that reads a recording and returns a verdict, so that the Python evaluator need not be executed per case. That goal is worth stating because it explains why the labels in every study come from a program (§3), and why the measurement failures took the particular shape they did. A programme whose ground truth is a program can only ever approach a tie, and five of the six instances are consequences of forgetting that in one way or another.

3 The setting

All six instances come from evaluating driving behaviour against rules. A rule is a predicate over a logged trajectory: a time-to-collision threshold, a comfort envelope, a lane-keeping condition. The reference implementation is a Python evaluator, so labels are exact and cheap, and the model's task is to reproduce a judgement that a program already makes. Four properties of this setting matter for reading the instances.

The labels come from a program. Every label in these studies is produced by the reference evaluator or by an exact certificate. This has a consequence that took us too long to state: the program defines the truth, so its score is perfect by construction and the model's ceiling is a tie. Any claim of the form "the model reaches accuracy x" is therefore a claim about how closely the model approximates a program that is already available, not about a capability the program lacks. Two of the six instances (§4) are consequences of forgetting this.

The base rate is high and known. Violations are rare in logged human driving. On the layer used for the central claim, the majority class covers 0.7031 of the rows (320 rows). An endpoint that aggregates over such a set is dominated by the majority class unless it is explicitly balanced.

Invalid output is a possible answer. A model can emit text that does not parse. Throughout, invalid and missing predictions count in the denominator as errors rather than being dropped, and the scorer asserts that the set of prediction identifiers equals the set of evaluation identifiers. Dropping unparsable rows silently converts a partial run into a strong one; this is a failure mode one of these studies hit before the assertion existed.

Arms, seeds and splits. An *arm* is a training recipe or a prompt condition. The studies compare arms seed by seed where training is involved, and report per-seed values rather than only a pooled difference. Splits are by log or by scenario, never by row, because rows from one log are not independent. Where a threshold is fitted, it is fitted on one set of scenarios and reported on another, with the scenario-level intersection asserted to be empty.

How the numbers here were produced. Every value is recomputed from a stored artifact by one generator script, which writes the inline macros and the tables together, so a sentence cannot disagree with the table beside it. Instance 1 is rescored from the stored prediction files with the shared scorer rather than copied from any summary. A separate checker re-runs the generator and fails if regeneration changes a byte, and scans the prose for bare numerals so that a hand-typed value cannot enter. While writing this paper that checker's precursor caught two values that the originating run log carried but that no stored artifact contained; §12 lists the substitution.

Table 1: The programme these instances come from. 5 studies ran over roughly a year, each with its own preregistration, frozen analysis script and held-out split. six instances are drawn from the 4 that produced quantitative endpoints; the fifth is listed because its conclusion shaped the others, not because it supplies a number. The mapping is by study, not by instance: the first study supplies three of the six.

Study	Question it asked	What it contributed here
Rule Execution and Supervision	Can a model learn to execute driving-evaluation rules, and does supervising the derivation beat supervising the verdict alone? Later: does outcome-only RL beat compute-matched continued supervision?	Instances 1, 2 and 5: the trivial upper bound, the training-seed noise floor, and the reversal when the reference arm got stronger.
Evidence Sufficiency	Under partial evidence, can a model tell whether a question is answerable at all?	Instances 3 and 6: the wording noise floor, and the readout gap between what the model knows and what it says.
World-Model Replay	Does a learned world model preserve the rule judgements the original recording supports?	Instance 4: a gate criterion that measured the data pipeline rather than the arm.
Runtime Rule Cards	Can one model be driven by a rule card supplied at inference time rather than trained in?	Instance 1b: a model-free executor of the program printed in the prompt scores perfectly.
Reasoner Comparison	Do general multimodal reasoners handle driving clips better than a frozen text baseline?	No measured instance ($n = 1$, qualitative). Its finding—that narrative fluency substituted for uncertainty, and that timestamps were produced that the frames did not support—is why uncertainty preservation is checked separately elsewhere.

Table 2: Instance 1. Every arm of the rule-execution study rescored from its stored predictions with the shared scorer. *Over* is accuracy minus the majority-class predictor on the same rows; it is negative for 4 of the 5 arms. Invalid and missing predictions count in the denominator.

Layer	Arm	Acc	Majority	Over	Bal. acc	Miss
L1	zero-shot	0.3809	0.9507	-0.5698	0.3979	0.5833
L1	result supervision	0.9877	0.9507	+0.0370	0.9891	0.0093
L2	zero-shot	0.6937	0.7031	-0.0094	0.4933	1.0000
L2	result supervision	0.6969	0.7031	-0.0062	0.7358	0.1684
L2	process supervision	0.6781	0.7031	-0.0250	0.7650	0.0211

4 Instance 1: a trivial upper bound on the metric

4.1 A majority-class predictor beats most of the arms

The study's central claim concerned a layer in which the model reads raw state and must compose rules it has not seen combined before. Three arms were run: zero-shot, supervision on the verdict alone, and supervision that also contained the derivation. Their accuracies were 0.6937, 0.6969 and 0.6781. Read on its own, the middle of those looks like a model that is right about seven times in ten.

The majority-class predictor on the same rows scores 0.7031. Subtracting it, the three arms are at -0.0094, -0.0062 and -0.0250: all three are below the constant predictor, and across both layers 4 of 5 rescored arms are. Table 2 gives the full rescoring.

The zero-shot arm makes the failure concrete. Its accuracy of 0.6937 is produced by answering the majority label on every one of 320 rows: its miss rate is 1.0000 and its balanced accuracy is 0.4933. There is no sense in which it has 0.6937 of the capability. It has none of it, and the number is an artifact of the base rate.

The easier layer shows why this is not simply a hard-task effect. There the base rate is higher still, 0.9507 over 2192 rows, and the supervised arm does clear it, reaching 0.9877 for a margin of +0.0370. That margin is small, but the balanced accuracy is 0.9891 and the miss rate 0.0093: on this layer the model really has the capability, and accuracy is simply a poor instrument for showing it. The same column that overstates the hard layer understates the easy one.

4.2 Three metrics, three orderings

The same three arms do not merely shift when the metric changes; they reorder. On accuracy, supervising the derivation is *worse* than supervising the verdict by -0.0188. On balanced accuracy it is better by +0.0292. On the miss rate—the quantity that matters if the judge is gating anything safety-relevant—it is better by a factor of 8.0.

A reader given only the accuracy column would conclude that adding the derivation hurt. A reader given only the miss rate would conclude it helped substantially. Both columns are correct. This is why the choice of primary endpoint has to be

fixed before the numbers exist, and why on a set with a known high base rate it should not be aggregate accuracy.

4.3 When the prompt contains the program

A second study in the same group produced a cleaner version of the same problem. Its prompt supplied the decision procedure explicitly, instructing the model to choose the status exactly as the enumerated steps dictate. The reported result was a perfect score, and successive iterations of the prompt had been credited with achieving it.

We wrote the trivial baseline that this framing implies: a short executor that reads the same facts and applies the same enumerated steps, with no model in the loop. It scores 24/24 on the synthetic suite of 24 cells and 12/12 on the 12 real cases, with 0 and 0 model calls respectively. The perfect score is therefore not evidence about reading a rule specification. It is evidence that the model transcribed a program it was handed, and the measurement cannot separate the two.

Two further details sharpen this. First, the executor is invariant to the order of the supplied rule cards by construction, because it reads facts by kind; and the entire prompt-iteration history of that study had been about card order and note formatting. What the iterations bought was robustness to layout, not rule semantics. Second, comparing the in-prompt program against the reference evaluator over their shared discrete input space shows 4 inputs on which the two disagree, because the reference has a branch the prompt never mentions. The number of real cases that reach that branch is zero, so the perfect agreement also does not establish that the program in the prompt is the program that defines the labels.

4.4 What would have caught it

Printing the majority-class predictor in the same table as the metric, and, when the prompt contains a procedure, executing that procedure as a baseline. Both are cheap. Neither was in the preregistration of either study.

5 Instance 2: a training-seed noise floor larger than the effect

The study compared arms seed by seed and reported, for each preregistered contrast, the difference at each of 2 training seeds over 8 held-out logs. That reporting choice is what made the instance visible; a pooled difference would have hidden it.

5.1 The same arm moves more than the arms differ

Holding the arm fixed and changing only the training seed moves the safety endpoint. Across the 6 arm-and-contrast combinations the stored analysis records, the within-arm movement in miss rate ranges from 0.0093 to 0.1574. The largest, 0.1574, belongs to a single arm on the layer where the study's first claim was made; on the harder layer the largest is 0.0909.

The consequence is visible in the contrasts themselves. 2 of the 3 preregistered contrasts change sign between the two seeds, namely H1 and H2. The first went from -0.0003 at one seed to +0.0309 at the other. Reported from either seed alone it would have supported a confident claim, in opposite directions.

The one contrast that keeps its sign is not therefore safe. Its two per-seed differences are -0.1484 and -0.0234, a factor of six apart, pooling to -0.0859 with a one-sided p of 0.9914. A stable sign with that much movement in magnitude still does not support an effect size.

5.2 The experiment had been sized against the wrong noise model

The study had done a power calibration before running, and had sized the sample from it. Its noise anchors were taken from discordance rates between 3 arm pairs expected to be near-equal: 0.0121, 0.0435, 0.0652, with a main correlation anchor of 0.08. Those anchors were estimated from scenario clustering, and they span 0.0121 to 0.0652.

They did not include training randomness. 3 of the 6 measured within-arm movements exceed 0.0652, the largest of the 3 anchors, and the largest movement, 0.1574, exceeds all of them and the correlation anchor as well. A calibration whose noise model omits a variance component that turns out to dominate will certify a sample size that cannot resolve the effect, which is what happened: the sample had been declared sufficient and was not.

5.3 Why this is not a null result

The tempting summary is that the compared recipes do not differ. That is not what the data say. The data say that at this number of seeds and this number of positive cases the comparison has no resolving power. The two statements lead to opposite decisions—one stops the line of work, the other adds seeds—so the distinction is worth keeping even though it is less quotable.

5.4 What would have caught it

Running the key contrast at more than one training seed before reporting it, and taking the noise anchor for any power calculation from same-configuration reruns rather than from cross-arm agreement. The second is the part that generalises: an anchor estimated from a source of variance that is not the one you will face is not a conservative choice, it is an unrelated one.

6 Instance 3: a wording noise floor with no training involved

The previous instance can be dismissed as a property of training. This one cannot. The arm here is zero-shot with greedy decoding, evaluated on 480 rows. Running the same prompt

Table 3: Instance 3. 7 semantics-preserving rewordings of one clause in a zero-shot evaluation; the interval sentence that follows it, the labels and the evidence are unchanged. *Over* is the primary endpoint minus the always-abstain baseline (0.5000). The span across rewordings is 0.0271, larger than the published arm's entire margin (+0.0250).

Causal clause of the constraint	Abstain	Over
the sensor reports only in *increments* of $0.5\,\mathrm{m/s}$	46	+0.0375
every speed is reported at a *resolution* of $0.5\,\mathrm{m/s}$	45	+0.0312
every speed is *quantised* to a step of $0.5\,\mathrm{m/s}$	42	+0.0292
every speed is *discretised onto a grid* of $0.5\,\mathrm{m/s}$	37	+0.0271
every speed is *rounded to the nearest* $0.5\,\mathrm{m/s}$	32	+0.0271
every speed carries a *measurement error* of up to $0.25\,\mathrm{m/s}$	28	+0.0167
each reported speed is *accurate to within* $0.25\,\mathrm{m/s}$	13	+0.0104

twice, on different shard counts and different hardware, produced byte-identical answers for every row. By the usual standard the measurement is perfectly reproducible and therefore noise-free.

6.1 Seven ways to say the same thing

The task requires the model to decide whether a question is answerable from the evidence given. One clause of the constraint block states that the reported speeds are coarse. We wrote 7 rewordings of that clause. The sentence that follows it, which gives the resulting interval explicitly, is identical to the character in all 7 variants; the labels are unchanged; the evidence is unchanged. Only the causal phrasing differs—whether the coarseness is described as quantisation, as resolution, as increments, or as a measurement error.

Table 3 gives the result. The primary endpoint's margin over the always-abstain baseline of 0.5000 moves between +0.0104 and +0.0375, a span of 0.0271. The margin the study had reported for its published wording is +0.0250, reached while abstaining on 44 of 480 rows. **The span across semantically equivalent rewordings is larger than the entire effect being reported.** The key behaviour moves with it: abstentions in the affected family of 96 rows range from 13 to 46, a spread of 33.

6.2 The spread is structured, not random

The orderings are not noise in the sense of being unpredictable. The discretisation words (quantised, resolution, increments, discretised, rounded) all produce high abstention; the error-tolerance words (accurate to within, measurement error) all produce low abstention. The wording decides whether the model frames the uncertainty as missing information or as a tolerance it can absorb, and that framing decides the answer.

Table 4: Instance 4. The gate criterion of the replay study, read from each arm's own summary. All 3 arms, including the constant-velocity control, report the same value to every stored digit, because the quantity is a property of the record-side projection rather than of the arm. Threshold 0.90.

Arm	Gate criterion	Passes
action-conditioned	0.8947	no
history-only	0.8947	no
constant velocity (control)	0.8947	no

This is a real finding about the model, and precisely for that reason it is not a property of the capability the endpoint was supposed to measure.

6.3 Reproducibility is not stability

The lesson we take is narrow and, we think, widely applicable: **byte-identical reruns at temperature zero measure implementation determinism, not measurement stability.** The noise floor of a zero-shot evaluation has to be estimated from a set of semantics-preserving rewordings. The same three conclusions from §5 then apply with "change the seed" replaced by "change the wording": report the band, declare effects inside it unresolved, and do not read reproducibility as low variance.

One honest qualification, which also appears in §12: these 7 rewordings were chosen by hand, and one of them had already been run when the others were written. The span is therefore an easily-reached range, not an interval estimate. It is enough to show the reported margin sits inside a wording band; it is not enough to quantify that band's width.

6.4 A second, smaller form of the same thing

The paraphrase sweep uses neutral evidence labels throughout, so its quantised cell is not the published arm but a variant that scores 0.5292 against the published 0.5250. The two prompts differ only in the label attached to an evidence row, not in the constraint text. We mention it because it is the kind of difference that is invisible in a results table and that we would have conflated had the generator not asserted the two were distinct.

7 Instance 4: a gate that measured the pipeline

The third study evaluated whether a learned world model, used to replay a scenario, preserves the rule judgements that the original recording supports. Its protocol put a gate before any comparison of arms: a set of agreement criteria that the pipeline had to pass for the downstream numbers to mean anything. Three arms were run, two of them models and the third a constant-velocity extrapolation included as a deliberately weak control.

Table 5: Instance 5. One question asked twice: is outcome-only RL better than continued supervised training under an equal wall-clock budget? The arms, the budget rule and the analysis are the same; the supervised starting point is stronger in the later round. The safety endpoint changes sign.

	weak start	strong start
RL miss	0.1761	0.0125
continued-SFT miss	0.5341	0.0109
favoured arm	RL	cont. SFT
seeds	2	10

7.1 All three arms fail at the same value

Every arm failed the same criterion, and failed it at the same number: the agreement between the record-side windowed projection and the representation ceiling is 0.8947 for all 3 arms, against a threshold of 0.90. The agreement is a count over logs—17 of 19—so the gate is decided by a single log.

Table 4 is deliberately dull: three rows with the same value. That identity is the content. The quantity is computed from the record-side projection and does not depend on which arm produced the replay, so it is not capable of distinguishing arms. Reading the failure as "the world model is not good enough" would have attributed a property of the data pipeline to the model.

7.2 Locating it changed the answer

Once the quantity was recognised as arm-independent, the cause could be looked for in the pipeline rather than in the model. It was the choice of kinematic estimator on the record side. With an estimator that takes the yaw from observed anchors on both sides, the agreement rises to 0.9474 and clears the threshold, and the event-level precision of the downstream comparison rises substantially. The related record-versus-reference agreement is 0.8421.

The general point is that the estimator is a free choice in the pipeline, and the gate verdict flips on it. A quantity that moves when a pipeline knob moves, and does not move when the model is replaced by a constant-velocity control, is not measuring the model.

7.3 What would have caught it

A single check, cheap enough to run on every criterion: substitute an obviously worse model and recompute. If the criterion does not change, it is not about the model. In this study the constant-velocity control was already there and already reported the identical value; what was missing was the step of comparing the control's value against the model arms' and noticing they were the same.

8 Instance 5: the reference arm decided the answer

The question is one that gets asked often: under an equal compute budget, does reinforcement learning on outcome-only rewards beat simply continuing supervised training? The same group asked it twice, roughly six months apart, with the same two arms, the same wall-clock budget rule and the same analysis machinery. The two rounds disagree in sign.

8.1 Round one: RL wins the safety endpoint decisively

In the first round the supervised starting point was weak. Under an equal budget, the outcome-only RL arm missed 0.1761 of the violations and the continued-supervised arm missed 0.5341. The safety-gate difference is -0.3785 with a confidence interval of [-0.4957, -0.2733], an interval that does not contain zero and a gap of nearly four tenths in miss rate. Read as a claim about RL, this is emphatic.

8.2 Round two: the same comparison reverses

The second round used a stronger recipe, and the supervised starting point it built on had been driven to a cost of 0.0102, far below the always-pass trivial baseline of 0.30. With 10 seeds instead of 2, the ordering reverses: the RL arm's cost is 0.0095 against 0.0075 for continued supervised training, a difference of +0.0020 in the wrong direction, with a one-sided p of 0.9863 and only 3 of 10 seeds favouring RL. The miss rate tells the same story: 0.0125 for RL against 0.0109 for continued training.

Table 5 puts the two rounds side by side.

8.3 The mechanism is reward saturation

This is not an unexplained reversal. When the supervised starting point is already strong, the outcome-only reward has nothing left to discriminate: in the calibration run the mean reward over the first ten steps was already high and the large majority of prompts had every sampled completion scoring identically, leaving no gradient. Under the identical wall clock the supervised arm took roughly five times as many optimisation steps as the RL arm, because each RL step costs a group of samples. The risk was written into the preregistration before the round was run, which is the only reason we can report it as a mechanism rather than a guess.

8.4 Why this instance is different from the first four

Instances 1 to 4 describe nuisance variables: quantities you can average over, bound, or hold fixed. More seeds, more rewordings, another estimator, a reported baseline. This one is not of that kind. "Outcome-only RL helps" has no truth value on its own; it has one only relative to a stated baseline strength.

Table 6: Instance 6. The same generation, scored two ways. The spoken endpoint is 0.5250 against an always-abstain baseline of 0.5000. Ranking the same decisions by the probability the model put on the token it emitted gives the coverage–risk numbers below, and thresholding it transfers to scenarios that do not appear in the fit (0 scenario overlap, asserted).

	AURC
random ordering (trivial)	0.4750
the model's own token probability	0.3392
oracle ordering (ceiling)	0.1372
	Accuracy
spoken endpoint on the held-out scenarios	0.5150
thresholded readout, fitted elsewhere	0.6650
refit on the held-out scenarios (ceiling)	0.6825

Averaging does not help, because there is nothing to average over—the comparison is well defined in each round and the rounds answer different questions.

The practical consequence is a reporting requirement rather than a design one: a comparison against a baseline must state where that baseline sits, on the same endpoint, against a trivial predictor. Our first round did not, and the resulting sentence is not wrong so much as incomplete in a way that invites the wrong generalisation.

8.5 An externally reported figure that points the same way

A recent driving-evaluation paper from another group reports, for its own setting, that reinforcement learning started from a zero-shot initialisation barely moves its validation accuracy, whereas the same procedure started from a supervised checkpoint improves it substantially [Sun et al., 2026]. We have not run their system and make no claim about their numbers; we use only the published values, and we note that their observation and ours bracket the same interval from opposite ends. Too weak a starting point and the search does not move; too strong and the reward saturates. If that is right, the usable window for outcome-only RL is narrower than either result alone suggests, and reporting the baseline's strength is what makes the two comparable at all.

9 Instance 6: the signal was there, the output was not

The five instances above all push the same way: a headline number that looked like capability was something else, and the honest correction is downward. This one is the reverse, and we think it is the most useful of the six. Here the headline number looked like absent capability, and that reading was also wrong.

9.1 The spoken endpoint is at the trivial baseline

The task is to decide, under partial evidence, whether a question is answerable at all. The model's spoken answer is correct on 0.5250 of 480 rows. Two constant strategies—always answer, always abstain—both score 0.5000. The margin is +0.0250. Taken alone, the natural conclusion is that the model cannot do this, and the natural next step is a different model or a different training objective.

9.2 The same generation, scored by its own probability

Instead of changing anything, we read out the probability the model assigned to the token it had already emitted at the decision position. Nothing else changed: the prompt, the contract and the decoding are the same, and the generated answers are identical to the published arm, which the analysis asserts rather than assumes. This matters: if the readout altered the answers it would be a different arm and the comparison would be void.

Ranking the 480 decisions by that probability and sweeping coverage gives an area under the risk–coverage curve of 0.3392. Random ordering gives 0.4750, which is also the full-coverage error rate; oracle ordering gives 0.1372. The model's own probability therefore captures 40.2% of the attainable gap between random and oracle, on decisions whose spoken form carries a margin of +0.0250.

9.3 It survives being turned into a rule and moved to new scenarios

A ranking is not yet a usable judge, so the next step was to make it a decision rule: abstain unless the probability exceeds a threshold. The threshold is fitted on 413 scenarios and reported on 321 others, with the scenario-level intersection asserted to be 0. On the 400 reporting rows, the rule reaches 0.6650 where the model's emitted token reaches 0.5150.

Three numbers are needed to read that honestly, and Table 6 carries all three. The trivial baseline on these rows is 0.5000, so the emitted token's own margin here is +0.0150. The rule's gain over the emitted token is +0.1500, which is 10 times that margin. And the ceiling obtained by refitting the threshold on the reporting scenarios themselves is 0.6825, so the transferred rule attains 90.4% of what refitting would have bought. Without that last number one cannot separate "a good threshold exists" from "the fitted threshold transfers". A single global threshold, rather than one per family, reaches 0.6125.

The threshold at one half reproduces the token the model emits exactly, so the original result is one member of this family of rules—the member that happens to sit at the least useful point.

9.4 Why this changes what to do next

The two diagnoses imply opposite actions. If the capability is absent, the response is a larger model or a new objective, and

it is expensive. If the capability is present but not connected to the output, the response is a decision rule or a calibration layer, and it is nearly free. Reading only the spoken endpoint cannot distinguish them, and in this case it pointed at the expensive one.

So the obligation is symmetric with §4. Before writing that a model lacks a capability, check with a readout that does not alter the task—token probability, an intermediate probe, self-consistency—whether the signal is present but unexpressed. "The output is empty, not the model" is not a figure of speech here; it is a difference of +0.1500 that transferred to scenarios the threshold had never seen.

10 What we do now

The six instances imply a small number of checks. We state them with their cost, because the reason none of them was in our preregistrations is not that they are hard but that they are easy to forget.

Print the trivial upper bound next to the metric. For a decision task, the majority-class predictor. For a task whose prompt contains a procedure, that procedure executed with no model. For a retrieval or tool condition, the same prompt without the tool—and if the two evidence payloads are byte-identical, the ablation does not exist. Cost: minutes. This one check accounts for two of the six instances.

On a skewed set, do not let aggregate accuracy be the primary endpoint. Fix the endpoint before the numbers exist, report the base rate in the same table, and for anything safety-relevant report the miss rate at a fixed false-alarm rate rather than each arm's own operating point. Two arms that each choose their own threshold are not comparable on their own miss rates.

Estimate the noise floor from the variance you will actually face. Training randomness if there is training; prompt wording if there is not. Byte-identical reruns do not bound either. Where an effect is inside the floor, write that the comparison does not resolve it at this size, which is a different statement from no effect and implies a different decision.

Test every gate criterion against a deliberately worse model. If the criterion does not move when the model is replaced by a weak control, it is measuring the pipeline. This is one line of analysis code and it was the whole of that instance.

State where the baseline sits. A comparison against a reference arm should report that arm's own distance from a trivial predictor on the same endpoint. Without it, a result that holds against a weak baseline will be read as a result about the method, and will reverse when the baseline improves.

Before concluding a capability is absent, try a readout that does not change the task. Token probability, an intermediate probe, self-consistency. Assert that the generated output is unchanged, or it is a different arm. Report the random and oracle orderings alongside any risk–coverage curve, and when a threshold is fitted, report the refit-on-the-reporting-set ceiling so that "a threshold exists" is distinguishable from "the threshold transfers".

Separate the three ways a threshold can be reported. Fitted and reported on the same rows; fitted elsewhere and reported here; refitted here as a ceiling. Split by scenario rather than by row, and assert the intersection is empty rather than arranging it and trusting the arrangement. A degenerate threshold imitates a capability, so report thresholds at full precision—a threshold at the boundary of its range is the trivial strategy wearing a number.

10.1 Packaging the first three

Three of our studies implemented the trivial-baseline computation separately, within the same week, and all three did so only after review. That is a signal about the component rather than about the people, so it is now one module: a scorer returning the metric together with the majority-class value and the margin over it, with invalid and missing predictions counted in the denominator rather than dropped, and with identifier coverage and denominator balance enforced by assertion rather than reported. Its acceptance test is that it reproduces the three hand-rolled results digit for digit, so the consolidation is checkable rather than asserted.

We do not present this as a contribution of substance. We present it because the alternative—a checklist in a document—had already failed in our own group three times in one week.

11 Where this leaves the research

A retrospective should say what it would do next. Each of the following follows from a specific instance above, and each is stated with the thing that would have to be built first.

From executing a known rule to understanding an unknown one. The headline result of the first study is capped by construction: the labels come from the Python evaluator, so a program scores perfectly and the model's ceiling is a tie. Every number in §4 is a measurement of how closely the model approximates a program that is already available. The questions with room in them are the ones the program cannot answer—whether a manoeuvre was reasonable given context, whether a rule applies at all in this scene—and those need labels that do not come from the evaluator. That makes annotation, not compute, the next bottleneck, and it is a different kind of project from the five described here. The external work we can read supports this: a recent benchmark reports a rule-based metric scoring near chance against human reasonable-

ness labels, which is headroom invisible to any study whose reference is the rule itself [Sun et al., 2026].

From a binary verdict to a confidence-aware judge. §9 is the strongest single lead we have. The spoken verdict carried a margin of +0.0150 over the trivial baseline; a threshold on the model's own token probability reached 0.6650 where the emitted token reached 0.5150, transferring to disjoint scenarios and attaining 90.4% of the refit ceiling. The judge we were building discarded most of what it knew at the moment of emitting a token. Building the confidence-aware version is cheap—a calibration layer and a frozen threshold, no retraining—and we have not done it. The obligation that comes with it is reporting: a selective judge has to be scored on a coverage–risk curve against random and oracle orderings, not on accuracy at full coverage.

From accuracy to safety-aware evaluation. On the hard layer, accuracy, balanced accuracy and miss rate ranked the same three arms in three different orders (§4): supervising the derivation was worse by -0.0188 on accuracy and better by a factor of 8.0 on misses. For a judge that gates anything, the miss rate at a fixed false-alarm rate is the quantity of interest, and arms that each pick their own operating point are not comparable on their own miss rates. We would now define the endpoint as a cost with the error ratio stated, as the later round did—while noting that the cost endpoint carries its own hostage, the assumed violation share of 0.10 against an observed 0.4467 (§12).

From running RL to first measuring whether there is room for it. The reversal in §8 was not bad luck. The mechanism—reward saturation at a strong supervised starting point, with the large majority of prompts having zero reward variance and the supervised arm taking roughly five times as many steps under the same wall clock—is measurable before the comparison is run, and in the second round it had been written into the preregistration as a risk. The cheap diagnostic is to sample the reward distribution at the intended starting point and check that it has variance to optimise. Run that first and the compute either gets spent on a question with an answer or gets saved.

From the model to the evaluation system. Four of the six instances were not about the model at all. One was a base rate (§4), one a program already present in the prompt, one a property of a record-side projection that read identically for a constant-velocity control (§7), and one a synonym (§6). Treating the judge as the object of study and the harness as plumbing is what let these through. The unit of analysis that would have caught them is the whole pipeline: data construction, label provenance, prompt rendering, decoding, parsing, denominator accounting, and the endpoint definition. §10 is a first pass at auditing that unit, and the one component we extracted—a scorer that reports the trivial upper bound beside the metric—was built only after three studies had hand-rolled it in the same week.

What we cannot currently do, and it matters. We would like to complement these instances with the same analysis applied to someone else's artifacts. That is presently not possible in this subfield. The rule-invoking judge of Sun et al. [2026] has released neither code nor data, and its source corpus is distributed under terms that designate benchmarking results relating to the dataset as confidential; the preference-based critic of Song et al. [2026] lists its code as pending release clearance. Re-reading published tables needs no access and is what we have done here. Independent re-measurement would need either a release or a benchmark built from openly licensed data, and the second is a more reliable plan than waiting for the first.

12 Limits of this evidence

Four limits are load-bearing: without them a number here would be read as more than it is. The rest of the caveats are in the sections that raise them.

Two values in the originating log could not be traced. The run log of the study in §5 quotes two within-arm miss-rate differences that appear in no stored artifact—only in that log's prose. We do not use them. The within-arm figures in §5 are recomputed from the per-seed block of the frozen analysis output, which is why they differ from the numbers that study's own log reports. This is the report's subject arising inside the report: a value written once in prose and then cited onward is not checkable, and ours were not. It is also why the build refuses to proceed if a section contains a hand-typed numeral.

One endpoint rests on an operational prior. The cost endpoint of §8 weights misses against false alarms using an assumed violation share of 0.10, fixed before the run. The observed share in the test set is 0.4467. Recomputing at the observed share leaves every direction unchanged but roughly doubles the absolute values, so the absolute costs should not be quoted outside the context of that prior.

The reversal compares rounds that differ in more than the baseline. The two rounds of §8 share the arms, the budget rule and the analysis machinery, but the second also changed the dataset construction, the rule families and the seed count, from 2 to 10. We attribute the reversal to the strength of the supervised starting point because the saturation mechanism was measured directly and preregistered as a risk. A contribution from the other changes cannot be ruled out; the clean version holds everything but the starting point fixed.

The rewordings are not a sample. The 7 rewordings of §6 were written by hand, and one had already been run when the rest were composed. The span of 0.0271 is the range that was easy to reach, not an interval estimate. Turning that instance into a quantitative claim needs a rewording procedure defined in advance.

Smaller notes. The replay study of §7 had not frozen its confirmatory run when these numbers were produced; the arm-independence of its gate criterion does not depend on that, but the effects of changing the estimator should be read as development findings. That gate agreement is 17 of 19 logs, so the comparison between 0.8947 and 0.9474 is one log changing hands rather than a precise effect size. And the six instances may be correlated through us: they span 4 studies, which shows they are not an artifact of one study, but not that they are not an artifact of one practitioner.

13 Related work

Judges and critics as evaluation components. Using a language or vision-language model to score or rank is now standard, and the surrounding literature has documented that such judges carry position, verbosity and self-preference biases [Zheng et al., 2023, Wang et al., 2024]. Those studies ask whether the judge is biased. The question here is adjacent and, we think, prior to it: whether the number used to certify the judge is measuring the judge at all. A position-bias correction does not help if the reported accuracy is below the majority-class predictor.

Trivial baselines and base rates. That a skewed test set makes aggregate accuracy uninformative is old and well understood [Provost et al., 1998]. The reason it recurs is not ignorance of the principle but the absence of the baseline from the table that gets read. Our contribution on this point is not the idea; it is the measurement of how far the principle's violation can travel—through preregistration and a frozen analysis—and the observation that in our group it was caught by review rather than by protocol in every case.

Variance and seed sensitivity. That fine-tuning outcomes vary substantially with random seed is established [Dodge et al., 2020], as is the broader point that reported gains often fall inside run-to-run variance [Henderson et al., 2018, Reimers and Gurevych, 2017]. §5 adds a specific and practical corollary: a power calculation whose noise anchor is estimated from a different variance component will certify a sample size that cannot resolve the effect, and will do so with the appearance of rigour. §6 extends the same logic to evaluations with no training, where the literature on prompt sensitivity has shown that formatting and phrasing move results substantially [Sclar et al., 2024, Lu et al., 2022]; our addition is the direct collision between a wording band and a reported margin in a setting that is byte-identical across reruns.

Selective prediction and calibration. Risk–coverage analysis and selective prediction have a long history [El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017], and token probability is a known if imperfect confidence signal in language models [Kadavath et al., 2022]. §9 uses these tools in a specific diagnostic role: distinguishing an absent capability from a capability that exists but is not expressed in the output, and then checking that the distinction survives being turned into a threshold and moved to disjoint scenarios.

Driving evaluation. Rule-based driving metrics are the dominant paradigm for open-loop benchmarking [Dauner et al., 2024, Cao et al., 2025], and recent work argues they are context-blind and proposes model-based judges in response. Song et al. [2026] train a preference-based critic on human-annotated trajectory pairs; Sun et al. [2026] instead have a vision-language model select which rule functions apply to a scene and aggregate only those, and report a large gain in AUC over the rule-based metric on their own human-labelled benchmark. We read those results as support for a claim we make only negatively here: the headroom that our studies could not find is headroom that appears once the reference is human judgement rather than a program, because a program-defined label caps the model at a tie (§3). Our relation to this work is confined to re-reading published tables; as §12 records, neither system's artifacts were available to us.

Measurement validity as a subject. Recent work has argued that machine-learning benchmarks should be examined as measurement instruments, with construct validity treated explicitly [Raji et al., 2021, Liao et al., 2021, Blodgett et al., 2021]. This retrospective is a small empirical addition to that programme: six cases where the construct and the number came apart, each with its mechanism identified and the cost of the missing check stated.

14 Conclusion

six times, across 4 studies, a headline number turned out to be decided by something other than the capability it was supposed to measure: a base rate, a program already printed in the prompt, a training seed, a choice of synonym, a property of the data pipeline, and the strength of the arm being compared against. The last ran the other way: an endpoint at the trivial baseline was read as absent capability while the model's own token probability ranked the same decisions at AURC 0.3392 against 0.4750 and transferred across disjoint scenarios.

Two things about this set seem worth keeping. First, the checks that would have caught each case are individually cheap—a majority-class column, a second seed, a reworded clause, a weak control substituted into a gate, a baseline's own distance from triviality, one readout that does not alter the task. None of them is a research contribution. All of them were missing from preregistrations that were otherwise careful, and four of the six were found by review rather than by protocol.

Second, the direction of the error is not consistent, and that is why a general scepticism is not an adequate response. Five of the instances mean a number was too high. One means a number was too low, and acting on the pessimistic reading would have been the more expensive mistake: it pointed at a new model when what was needed was a threshold. A practice that only subtracts credulity will get that case wrong too.

What this evidence does not support is any claim about whether models can judge driving. Every instance is about a measurement. We also cannot say how common these failures are in the literature, and in this particular subfield we are currently unable to find out: the comparable recent systems have not released code or data, and one of them rests on a corpus whose terms treat benchmarking results as confidential. The most we can do from outside is re-read published tables, which is how we noticed that the trivial baseline for one such table is the majority class at a value its own random-ordering row does not report.

The uncomfortable summary is that none of this was discovered by being careful in advance. It was discovered by someone asking, after the number was written down, what else could have produced it. We have turned the six answers into a checklist and a scorer, which is a way of not having to remember. Whether that is enough we will find out the next time a reviewer asks.

References

Su Lin Blodgett, Gilsinia Lopez, Alexandra Olteanu, Robert Sim, and Hanna Wallach. Stereotyping norwegian salmon: An inventory of pitfalls in fairness benchmark datasets. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, 2021.

Wei Cao, Marcel Hallgarten, Tianyu Li, Daniel Dauner, Xunjiang Gu, Caojun Wang, Yakov Miron, Marco Aiello, Hongyang Li, Igor Gilitschenski, Boris Ivanovic, Marco Pavone, Andreas Geiger, and Kashyap Chitta. Pseudo-simulation for autonomous driving. In *Conference on Robot Learning*, 2025.

Daniel Dauner, Marcel Hallgarten, Tianyu Li, Xinshuo Weng, Zhiyu Huang, Zetong Yang, Hongyang Li, Igor Gilitschenski, Boris Ivanovic, Marco Pavone, Andreas Geiger, and Kashyap Chitta. NAVSIM: Data-driven non-reactive autonomous vehicle simulation and benchmarking. In *Advances in Neural Information Processing Systems*, volume 37, pages 28706–28719, 2024.

Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah A. Smith. Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping. *arXiv preprint arXiv:2002.06305*, 2020.

Ran El-Yaniv and Yair Wiener. On the foundations of noise-free selective classification. *Journal of Machine Learning Research*, 11: 1605–1641, 2010.

Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems 30*, 2017.

Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, 2018.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Thomas I. Liao, Rohan Taori, Inioluwa Deborah Raji, and Ludwig Schmidt. Are we learning yet? a meta review of evaluation failures across machine learning. In *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*, 2021.

Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, 2022.

Foster J. Provost, Tom Fawcett, and Ron Kohavi. The case against accuracy estimation for comparing induction algorithms. In *Proceedings of the Fifteenth International Conference on Machine Learning*, 1998.

Inioluwa Deborah Raji, Emily M. Bender, Amandalynne Paullada, Emily Denton, and Alex Hanna. AI and the everything in the whole wide world benchmark. In *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*, 2021.

Nils Reimers and Iryna Gurevych. Reporting score distributions makes a difference: Performance study of LSTM-networks for sequence tagging. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, 2017.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design, or: How i learned to start worrying about prompt formatting. In *The Twelfth International Conference on Learning Representations*, 2024.

Jingyu Song, Zhenxin Li, Shiyi Lan, Xinglong Sun, Nadine Chang, Maying Shen, Joshua Chen, Katherine A. Skinner, and Jose M. Alvarez. DriveCritic: Towards context-aware, human-aligned evaluation for autonomous driving with vision-language models. In *IEEE International Conference on Robotics and Automation*, 2026. arXiv:2510.13108.

Xinglong Sun, Kevin Xie, Jenny Schmalfuss, Despoina Paschalidou, Xiuming Zhang, Sanja Fidler, Kashyap Chitta, and Jose M. Alvarez. DriveJudge: Rethinking autonomous driving evaluation with vision-language models. *arXiv preprint arXiv:2606.17362*, 2026.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems 36: Datasets and Benchmarks Track*, 2023.